La Scuola d'Italia / Dr. Huson / IB Math: Practice
29 April 2026

First & Last name:

11.15 Classwork: Periodic Function Modeling (calculator)

1. [Maximum mark: 7]

   A particle moves along a straight line so that its velocity, v m s^{-1}, after t seconds is given by

   $$v(t) = e^{\sin t} + 4\sin t, \text{ for } 0 \le t \le 6.$$

   (a) Find the value of t when the particle is at rest. [2]

   (b) Find the acceleration of the particle when it changes direction. [3]

   (c) Find the total distance travelled by the particle. [2]

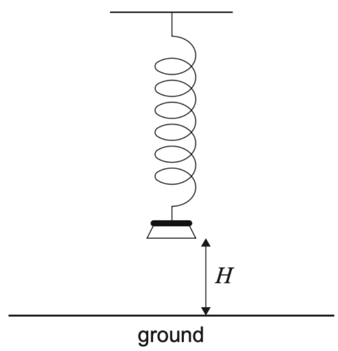
2. [Maximum mark: 13]

A weight suspended on a spring is pulled down and released, so that it moves up and down vertically.

The height, H metres, of the base of the weight above the ground can be modelled by the function

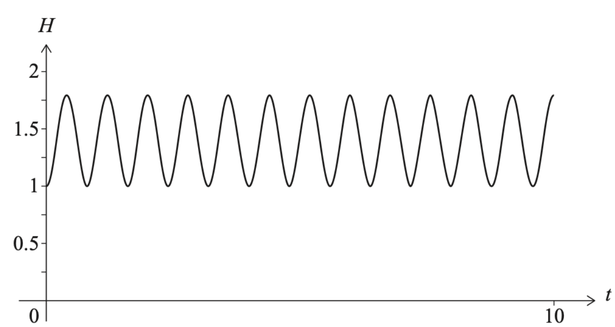
$$H(t) = a\cos(7.8\,t) + b, \quad a, b \in \mathbb{R}, \quad 0 \le t \le 10,$$

where t is the time in seconds after the weight is released.



(a) Find the period of the function. [2]

The weight is released when its base is at a minimum height of 1 metre above the ground, and it reaches a maximum height of 1.8 metres above the ground. The graph of H is shown in the following diagram.



(b) Find the value of (i) a; (ii) b. [3]

(c) Find the number of times that the weight reaches its maximum height in the first five seconds of its motion. [2]

(d) Find the first time that the base of the weight reaches a height of 1.5 metres. [2]

A camera is set to take a picture of the weight at a random time during the first five seconds of its motion.

(e) Find the probability that the height of the base of the weight is greater than 1.5 metres at the time the picture is taken. [4]

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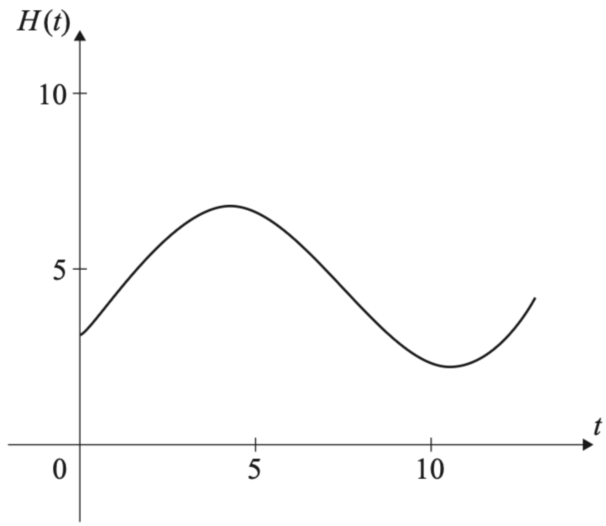
3. [Maximum mark: 13]

The height of water, in metres, in Dungeness harbour is modelled by the function

$$H(t) = a\sin(b(t-c)) + d,$$

where t is the number of hours after midnight, and a, b, c, d are constants with $a > 0$, $b > 0$ and $c > 0$.

The following graph shows the height of the water for 13 hours, starting at midnight.



The first high tide occurs at 04:30 and the next high tide occurs 12 hours later. Throughout the day, the height of the water fluctuates between 2.2 m and 6.8 m. All heights are given correct to one decimal place.

(a) Show that $b = \dfrac{\pi}{6}$. [1]

(b) Find the value of a. [2]

(c) Find the value of d. [2]

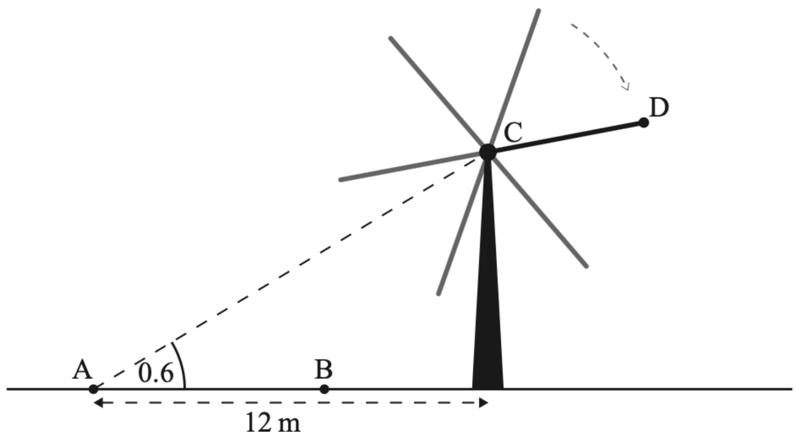
(d) Find the smallest possible value of c. [3]

(e) Find the height of the water at 12:00. [2]

(f) Determine the number of hours, over a 24-hour period, for which the tide is higher than 5 metres. [3]

4. [Maximum mark: 13]

The six blades of a windmill rotate around a centre point C. Points A and B and the base of the windmill are on level ground, as shown in the following diagram.



From point A the angle of elevation of point C is 0.6 radians.

(a) Given that point A is 12 metres from the base of the windmill, find the height of point C above the ground. [2]

An observer walks 7 metres from point A to point B.

(b) Find the angle of elevation of point C from point B. [2]

The observer keeps walking until he is standing directly under point C. The observer has a height of 1.8 metres, and as the blades of the windmill rotate, the end of each blade passes 2.5 metres over his head.

(c) Find the length of each blade of the windmill. [2]

One of the blades is painted a different colour than the others. The end of this blade is labelled point D. The height h, in metres, of point D above the ground can be modelled by the function

$$h(t) = p\cos\left(\frac{3\pi}{10}t\right) + q,$$

where t is in seconds and $p, q \in \mathbb{R}$. When $t = 0$, point D is at its maximum height.

(d) Find the value of p and the value of q. [4]

If the observer stands directly under point C for one minute, point D will pass over his head n times.

(e) Find the value of n. [3]